No. 13, Delma Street, Al Nahyan Area
Abu Dhabi, United Arab Emirates
gaoclarie@gmail.com

Webpage : https://rrgao.github.io
Github :
+971-58-5767960

Rongrong GAO

AI PhD and Researcher, HKUST

Objective

I have joined **SenseTime** in 2017 and **Allybot** in 2018, respectively, where I have been working on AI research and device-level development in ***intelligent AI systems*** and ***autonomous driving and robotics perception systems***. From 2019 to 2024, I continued my Ph.D. career at the Hong Kong University of Science and Technology (**HKUST**), focusing on AI-related research in ***3D Vision***, ***multi-modal AI*** and ***GenAI*** applications.
Adept at general AI, I am now eager to seek a challenging engineering position to contribute my expertise and drive organizational success.

Education

Ph.D., Computer Science and Engineering *Sep' 2019 - Oct' 2024*
Dep. of CSE, Hong Kong University of Science and Technology , Hong Kong
Major in 3D robotics perception systems and GenAI, under the supervision of Prof. Qifeng Chen.

Professional Experience

GenAI and 3D representations - AI Startup, UAE *Nov' 2024 - Present*

Developed a real-time multi-modal-to-3D generation system that converts user inputs from multiple modalities into high-fidelity 3D representations on embedded, low-latency hardware platforms.
Designed and optimized efficient 3D reconstruction and generation pipelines for resource-constrained edge devices. Leveraged generative AI and 3D representation learning to support digital content ownership protection and commercialization for artists and creators.

AI perception systems for robotics - HKUST, Hong Kong *Sep' 2019 - Oct' 2024*

Watermarking for Generative Foundation Models

- Proposed a novel training-data copyright protection framework for large-scale generative models.
- Designed differentiable watermark embedding and extraction networks robust to diffusion sampling, model fine-tuning, and common post-processing operations by formulating the task as an information-theoretic optimization problem, enabling imperceptible yet reliable ownership verification.
- Investigated copyright protection mechanisms for foundation models and generative AI systems.

Multi-modal Representation Learning for RGB-D Data

- Developed a cross-modal information hiding framework that embeds depth information into RGB images and reconstructs it with high fidelity.
- Designed a multi-modal feature fusion architecture for joint representation learning by exploiting cross-modality redundancy between RGB and depth data through feature disentanglement and representation learning.
- Addressed challenges in storage, transmission of diverse multi-modal data.

3D Vision, Geometric Learning, and Scene Understanding

- Developed the Semantics-and-Geometry-aware Network (SGNet) for point cloud colorization, jointly modeling geometric structure and semantic context.
- Constructed the ToF-100 dataset and designed a joint depth-normal estimation framework for geometric scene understanding.
- Built an end-to-end sensing-to-learning pipeline integrating data acquisition, annotation, model training, and evaluation for real-world robotic perception applications.

Autonomous driving perception system - Allybot, China *June'2018 - Sep'2019*

- Developed real-time perception algorithms for autonomous driving, including object detection, multi-object tracking, semantic segmentation, and 3D localization of vehicles, pedestrians, and traffic infrastructure.
- Built a large-scale camera-LiDAR dataset and established a complete data pipeline covering collection, annotation, quality control, and dataset construction.
- Designed an image–point cloud fusion framework to improve perception robustness under adverse weather and challenging environmental conditions.
- Architected and integrated the perception software stack with localization, planning, and control modules within the autonomous driving system.
- Optimized and deployed deep learning models using CUDA, TensorRT, and ONNX for real-time on-board inference.
- Collaborated on system validation, code optimization, and software version management, improving reliability and operational stability through both simulation and real-world testing.

Selected Publications

FIRST and CO-AUTHOR:
(* Corresponding author; † equal contributions.)

- B Liu, Z Huang, Y Li, **R Gao**, HX Chen, TZ Xiang.
  HATFormer: Height-aware Transformer for multi-modal 3D change detection.
  ISPRS Journal of Photogrammetry and Remote Sensing (ISPRS), 2025.
- R Wu, TZ Xiang, GS Xie, **R Gao**, X Shu, F Zhao, L Shao.
  Uncertainty-aware transformer for referring camouflaged object detection.
  IEEE Transactions on Image Processing (TIP),2025.
- ZJ Wu, **R Gao**, TZ Xiang.
  Scribble-Based Weakly Supervised Camouflaged Object Detection via SAM-Guided Feature Correlation Transformer.
  European Conference on Artificial Intelligence (ECAI), 2025.
- B Dong, J Pei, **R Gao**, T-Z Xiang*, S Wang, H Xiong.
  A Unified Query-based Paradigm for Camouflaged Instance Segmentation.
  ACM International Conference on Multimedia (ACM MM), 2023.
- Z Chen, **R Gao**, T-Z Xiang, F Lin.
  Diffusion Model for Camouflaged Object Detection.
  European Conference on Artificial Intelligence (ECAI), 2023.
- R Gao, **T-Z Xiang**, C Lei, J Park, Q Chen.
  Scene-level Point Cloud Colorization with Semantics-and-geometry-aware Networks.
  IEEE International Conference on Robotics and Automation (ICRA), 2023.
- **R Gao**, T-Z Xiang, C Lei, J Park, Q Chen.
  Scene-level Point Cloud Colorization with Semantics-and-geometry-aware Networks.
  IEEE International Conference on Robotics and Automation (ICRA), 2023.
- **R Gao**, N Fan, C Li, W Liu, Q Chen.
  Joint depth and normal estimation from real-world time-of-flight raw data.
  IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2021.
- Y Wu, **R Gao**, J Park, Q Chen.
  Future video synthesis with object motion prediction.
  Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern (CVPR), 2020.

Honors & Awards

2019–2024 Hong Kong PhD Fellowship Scheme (HKPFS),	Top 8%
2016 National Scholarships of China, Wuhan University,	Top 1%
2014 Outstanding Graduates, Wuhan University,	Top 5%
2013 National Scholarships of China, Wuhan University,	Top 5%
2012 National Scholarships of China, Wuhan University,	Top 1%
2011 National Scholarships of China, Wuhan University,	Top 1%

Computer Skills

Programming Languages: Python, MATLAB, C/C++, LaTeX
Research Tools: PyTorch, TensorFlow, OpenCV, Docker
Expertise: Computer Vision, Remote Sensing, Photogrammetry, Deep Learning